CORDA Algorithm

System Initialization & Constants

- N → Number of UAV
- k_{cable} → Cable Stifness
- m_P → Payload Mass
- m_l → Liquid Mass
- $m_{tot} = m_P + m_l$ → Total Mass
- $\vec{GC} = [0, 0, 0]$ → Geometric Center
- $\mathbf{I}_{GC}$ → Payload Inertial Tensor relative to GC
- $\vec{CoM} = [x_{CoM}, y_{CoM}, z_{CoM}]$ → Center of Mass
- $\vec{e}_{static}$ → Static Structural Eccentricity
- $\vec{e}_{slosh}(t)$ → Dynamic Displacement of the Liquid Centroid
- $\vec{e}_{CoM} = \vec{e}_{static} + \frac{m_l}{m_{tot}}\vec{e}_{slosh}$ → CoM Eccentricity
- $\vec{CoP} = [x_{CoP}, y_{CoP}, z_{CoP}]$ → Center of Pressure relative to GC
- $\vec{CoP}_{eff} = \vec{CoP} - \vec{CoM}$ → Center of Pressure relative to CoM
- $\mathbf{I}_{corr} = m_{tot}(\|e_{\vec{CoM}}\|^2\mathbf{I}_3 - e_{\vec{CoM}}{e_{\vec{CoM}}}^T)$ → Payload Inertial Tensor Correction
- $\mathbf{I}_{CoM} = \mathbf{I}_{GC} - k_{I,offset}\mathbf{I}_{corr}$ → Payload Inertial Tensor relative to CoM
- $\vec{u}_{rad,i} = [\cos(\frac{2\pi(i-1)}{N}), \sin(\frac{2\pi(i-1)}{N})]^T$ → Flat Radial Unit Vector
- $\vec{att}_i = [x_{att,i}, y_{att,i}, z_{att,i}]$ → Coordinates of Attachment Points relative to GC
- $\vec{att}_{eff,i} = \vec{att}_i - \vec{CoM}$ → Coordinates of Attachment Points relative to CoM
- L_{des} → Desired Cable Length.
- $F_{nom} = \max(m_{tot}g, 1.0)$ → Nominal Force for Normalization
- $M_{nom} = F_{nom} \cdot (L_{payload}/2)$ → Nominal Moment for Normalization

State Estimation & Inputs

- $\vec{a}_{des}$ → Desired Payload Acceleration
- $\vec{\omega}_p, \dot{\vec{\omega}}_{des}$ → Payload Angular Velocity & Desired Angular Acc.
- v_{wind} → Wind Velocity
- $v_{rel} = v_p - v_{wind}$ → Relative Airspeed Vector
- $\vec{F}_{aero,P} = -\frac{1}{2}\rho C_{d,p} A_{proj} \|v_{rel}\|^2 \hat{u}_{rel}$ → Wind Force
- $\vec{F}_{inertial} = -m_{tot} \cdot \vec{a}_{cmd}$ → Inertial Force
- $\vec{F}_{int} = -m_{tot} \cdot \ddot{\vec{e}}_{cmd}$ → Internal Force
- $\vec{F}_{ext} = \vec{F}_{aero,P} + \vec{F}_{inertial}$ → External Forces
- $\vec{F}_{tot} = -m_{tot}\vec{\mathbf{g}} + \vec{F}_{ext} + \vec{F}_{int}$ → Total Force Vector
- $\vec{F}_{req} = -\vec{F}_{tot}$ → Required Force Vector
  - $F_{req,z} = \max(F_{req,z},\ 0.1)$
- $\hat{u}_{F_{req}} = \frac{\vec{F}_{req}}{\|\vec{F}_{req}\|} + \epsilon$ → Force Direction Unit Vector

Yaw Logic

- β_{force} → Force Low-Pass Filter Constant
- $F_{yaw,min}, F_{yaw,max}$ → Weather Vane Force Thresholds
- $\vec{F}^{(k)}_{xy,filt} = (1-\beta_{force})\vec{F}^{(k-1)}_{xy,filt} + \beta_{force}\vec{F}_{req,xy}$ → Filtered Required Force on xy plane
- $\psi_{target,raw} = \mathrm{atan2}(F^{(k)}_{y,filt}, F^{(k)}_{x,filt})$ → Optimal Yaw Giving disturbances
- $\psi_{target} = \psi_{target,raw} + \psi_{offset} + \Delta\psi_{lag}$
  - $\psi_{offset,sym} = \frac{\pi}{N}$ → Offset for Symmetric Payload
  - $\psi_{offset,rect} = \begin{cases} 0 & \text{if wide} < \text{length} \\ \frac{\pi}{2} & \text{otherwise} \end{cases}$ → Offset for Rectangular Payload

- $\eta_\psi = \text{clip}\left(\frac{\|\vec{F}^{(k)}_{xy,filt}\| - F_{yaw,min}}{F_{yaw,max} - F_{yaw,min}}, 0, 1\right)$ → Yaw Autority
- $\psi_{err,raw} = \text{normalize_angle}(\psi_{target} - \psi^{(k-1)}_{stable})$
- $\psi_{err,eff} = \begin{cases} 0 & \text{if } |\psi_{err,raw}| < \psi_{db} \\ \psi_{err,raw} - \text{sgn}(\psi_{err,raw}) \cdot \psi_{db} & \text{otherwise} \end{cases}$
- $\Delta\psi = \text{clip}(\eta_\psi K_\psi \psi_{err,eff} + \dot{\psi}_{target}\Delta t, -\Delta\psi_{max}, \Delta\psi_{max})$
- $\psi_{eff} = \psi^{(k-1)}_{stable} + \Delta\psi$ → Effective Yaw used for Rotation Matrices

Attitude Logic

- λ_{aero} → Aerodynamic Blending Factor
- $\vec{F}_{req,\psi} = \mathbf{R}_z(\psi_{target})^T \cdot \vec{F}_{req}$ → Yaw-Despun Force Vector (removes Yaw rotation to find correct Roll/Pitch)
- $\theta_{aero} = \arctan 2(F_{req,\psi,x}, F_{req,\psi,z})$ → Aerodynamic Pitch (Aligns X-axis with force)
- $\phi_{aero} = \arctan 2(-F_{req,\psi,y}, \sqrt{F^2_{req,\psi,x} + F^2_{req,\psi,z}})$ → Aerodynamic Roll (Aligns Y-axis with force)
- $\phi_{blend} = (1 - \lambda_{aero}) \cdot \phi_{des,p} + \lambda_{aero} \cdot \phi_{aero}$ → Blended Roll
- $\theta_{blend} = (1 - \lambda_{aero}) \cdot \theta_{des,p} + \lambda_{aero} \cdot \theta_{aero}$ → Blended Pitch

Slosh Filter & CoM Correction

- τ_{slosh} → Sloshing Filter Time Constant
- $\beta_{slosh} = \frac{\Delta t}{\tau_{slosh} + \Delta t}$ → Slosh Filter Coefficient
- $\vec{e}_{CoM,\psi} = \mathbf{R}_z(\psi_{target})^T \cdot \vec{e}_{CoM}$ → Yaw-Despun CoM Eccentricity
- $\vec{e}_{CoM,\psi,filt,k} = (1 - \beta_{slosh}) \cdot \vec{e}_{CoM,\psi,filt,k-1} + \beta_{slosh} \cdot \vec{e}_{CoM,\psi}$
- $\phi_{CoM} = \arctan\left(\frac{\vec{e}_{CoM,\psi,filt,y}}{L_{des}}\right)$ → Balanced Roll

- $\theta_{CoM} = -\arctan\left(\frac{\vec{e}_{CoM,\psi,filt,x}}{L_{des}}\right)$ → Balanced Pitch

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- λ_{CoM} → Stability Blending Factor
- $\phi_{final} = k_{tilt} \cdot \phi_{blend} + \lambda_{CoM} \cdot \phi_{CoM}$ → Final Roll
- $\theta_{final} = k_{tilt} \cdot \theta_{blend} + \lambda_{CoM} \cdot \theta_{CoM}$ → Final Pitch
- $\mathbf{R}_{des,p} = \mathbf{R}_z(\psi_{target}) \cdot \mathbf{R}_y(\theta_{blend}) \cdot \mathbf{R}_x(\phi_{blend})$ → Blended Payload Attitude
- $\mathbf{R}_{des,f} = \mathbf{R}_z(\psi_{target}) \cdot \mathbf{R}_y(\theta_{final}) \cdot \mathbf{R}_x(\phi_{final})$ → → Blended Formation Attitude
- $\mathbf{R}_{tilt} = \mathbf{R}_{des,f} \cdot \mathbf{R}_z(\psi_{target})^T$ → Pure Tilt Rotation Matrix (yawed only roll and pitch)

Moment & Shifts Requirements

- $\vec{M}_{inertial} = \mathbf{R}_{des,p}(\mathbf{I}_{CoM}\dot{\vec{\omega}}_{des} + \vec{\omega}_p \times (\mathbf{I}_{CoM}\vec{\omega}_p))$
- $\vec{M}_{slosh} = \mathbf{R}_{des,p}(\vec{e}_{slosh} - \vec{e}_{CoM}) \times (\mathbf{R}_{des,p}F_{int})$
- $\vec{M}_{aero} = (\mathbf{R}_{des,p}\vec{CoP}_{eff}) \times \vec{F}_{aero}$
- $\vec{M}_{coriolis} = \mathbf{R}_{des,p}(\vec{e}_{slosh} - \vec{e}_{CoM}) \times (2m_l\vec{\omega}_p \times (\mathbf{R}_{des,p}\dot{\vec{e}}_{CoM}))$
- $\vec{M}_{req} = \vec{M}_{inertial} - \vec{M}_{slosh} - \vec{M}_{aero} - \vec{M}_{coriolis}$
- $\vec{W}_{req} = \begin{bmatrix} \vec{F}_{req} \\ \vec{M}_{req} \end{bmatrix}$ → Required Wrench

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- λ_{static} → CoM Static Blending Factor
- $\vec{\Delta}_{static} = \vec{e}_{CoM,\psi,filt} \cdot \lambda_{static}$ → CoM Static Shift Vector

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- β_{mom} → Momentum Shift Filter Constant
- $\vec{\Delta}_{mom,raw} = \begin{cases} \frac{\vec{F}_{req} \times \vec{M}_{req}}{\|\vec{F}_{req}\|^2} & \text{if } \|\vec{F}_{req}\|^2 > 10^{-2} \\ [0,0,0]^T & \text{otherwise} \end{cases}$ → Momentum Shift Vector

- $\vec{\Delta}_{mom}^{(k)} = \beta_{mom} \cdot \vec{\Delta}_{mom,raw} + (1 - \beta_{mom}) \cdot \vec{\Delta}_{mom}^{(k-1)}$ → Filtered Momentum Shift Vector

Shape Transformation (Ellipse)

- λ_{shape} → Shape Blending Factor
- $\hat{u}_{F_{req}}^{\perp} = \begin{bmatrix} -F_{req,y} \\ F_{req,x} \end{bmatrix} \frac{1}{\|\vec{F}_{req,xy}\|}$ → Required Forces perpendicular direction versor
- $\mathbf{R}_{req} = \left[\hat{u}_{F_{req}}, \hat{u}_{F_{req}}^{\perp}\right]$ → Required Force Direction Basis
- $S_{fact} = \min\left(\frac{\|\vec{F}_{req,xy}\|}{F_{ref}}, 2.0\right)$ → Scale Factor depending from Required Forces ($F_{ref} \approx 15$)
- $k_{\|} = \max\left(1.0 - (1.0 - \lambda_{par}) \cdot \min(S_{fact}, 1.0),\ 0.6\right)$ → Ellipse Scale Factor Parallel to Wind direction ($\lambda_{par} \approx 0.8$)
- $k_{\perp} = 1.0 + (\lambda_{perp} - 1.0) \cdot S_{fact}$ → Ellipse Scale Factor Perpendicular to Wind direction ($\lambda_{perp} \approx 1.8$)
- $\mathbf{S}_{def} = \begin{bmatrix} k_{\|} & 0 \\ 0 & k_{\perp} \end{bmatrix}$ → Deformation Matrix
- $\mathbf{S}_{mixed} = \mathbf{I}_2 \cdot (1 - \lambda_{shape}) + \mathbf{S}_{def} \cdot \lambda_{shape}$
- $\mathbf{M}_{shape} = \mathbf{R}_{req} \cdot \mathbf{S}_{mixed} \cdot \mathbf{R}_{req}^T$ → Shape Transformation Matrix
- $\vec{u}_{def,i} = \mathbf{M}_{shape} \cdot \vec{u}_{rad,i}$ → Deformed 2D Radial Vector

Parameters for Optimization

- λ_{twist} → Twist Gain Factor

- $\vec{T}_{safe} = k_{safe} \cdot \left(\frac{m_{tot} \cdot \vec{g}}{N}\right) + k_{safe2} \frac{\|\vec{F}_{ext}\|}{N}$ → Safe Tension ($k_{safe} \approx 0.1$)
- $\vec{T}_{limit} = (\vec{F}_{max,thrust} - m_{drone} \cdot \vec{g}) \cdot k_{limit}$ → Limit Tension ($k_{limit} \approx 0.7$)

- $\alpha_{dyn} = \arctan\left(\frac{\|\vec{F}_{req,xy}\|}{|\vec{F}_{req,z}|}\right)$ → Minimum angle to prevent slack due to external forces. If $\alpha < \alpha_{dyn}$ cable slack occurs and geometric control is lost.
- α_{min} → Minimum Desired Alpha
- $\alpha_{safe} = \max(1.1 \cdot \alpha_{dyn}, \alpha_{min})$
- $\alpha_{limit} = 0.9 \cdot \frac{\pi}{2}$
- $\alpha_{prev} = \alpha_{opt}^{(k-1)}$
- δ_{max} → Max Angle Variation per step
- $\alpha^*_{safe} = \max(\alpha_{safe}, \alpha_{prev} - \delta_{max})$
- $\alpha^*_{limit} = \min(\alpha_{limit}, \alpha_{prev} + \delta_{max})$

- α^2_{scale} → to normalize angular error
- α^2_{smooth} → to normalize smoothing
- $cond_{scale}$ → to scale the condition number

- α_{ref} → Desired Alpha
- $k_{\theta,boost}$ → Cone Angle Boost Gain
- $\alpha_{boost} = k_{\alpha,boost} \cdot \text{clip}\left(\frac{\|\vec{M}_{req}\|}{M_{nom,rot}}, 0, 1\right)$
- $\alpha_{ref,dyn} = \alpha_{ref} + \alpha_{boost}$

Optimization Loop (Find α_{opt})

Objective: $\alpha_{opt} = \underset{\alpha_k \in [\alpha^*_{safe}, \alpha^*_{limit}]}{\text{argmin}} \; J(\alpha_k)$

Inside the Loop

1. $h(\alpha_k) = L_{des}\cos(\alpha_k)$ → Vertical height relative to CoM
2. $r(\alpha_k) = h(\alpha_k)\tan(\alpha_k)$ → Horizontal radius of the drone formation
3. **Twist:**

a. $f_{z,safe} = \max(|W_{req,F_z}|, 1.0)$

b. $r_{safe}(\alpha_k) = \max(r(\alpha_k), 0.1)$

c. $\gamma_{raw}(\alpha_k) = \lambda_{twist} \cdot \frac{\tilde{M}_{req,z}}{f_{z,safe} \cdot r_{safe}(\alpha_k)}$

d. $\gamma_{twist}(\alpha_k) = \text{clip}(\gamma_{raw}(\alpha_k), -45°, 45°)$

e. $\mathbf{R}_{twist}(\alpha_k) = \begin{bmatrix} \cos\gamma_{twist}(\alpha_k) & -\sin\gamma_{twist}(\alpha_k) \\ \sin\gamma_{twist}(\alpha_k) & \cos\gamma_{twist}(\alpha_k) \end{bmatrix}$

4. $\vec{\Delta}_{shift}(\alpha_k) = h(\alpha_k) \cdot \frac{\|\vec{F}_{req,xy}\|}{|\vec{F}_{req,z}|}$ → Wind Shift

5. **Drone Position:**

a. $\vec{u}_{rad,\psi,i} = \mathbf{R}_z(\psi_{target}) \cdot \vec{u}_{rad,i}$

b. $\vec{u}_{rad,\psi,def,i} = \mathbf{M}_{shape} \cdot \vec{u}_{rad,\psi,xy,i}$

c. $\vec{u}_{rad,\psi,def,\gamma,i} = \mathbf{R}_{twist}(\alpha_k) \cdot \vec{u}_{rad,\psi,def,i}$

d. $\vec{\mathbf{D}}_i(\alpha_k) = \vec{CoM} + \mathbf{R}_{des,f} \cdot \vec{att}_{eff,i} + \mathbf{R}_{tilt} \cdot \begin{bmatrix} r(\alpha_k)\vec{u}_{rad,\psi,def,\gamma,x,i} \\ r(\alpha_k)\vec{u}_{rad,\psi,def,\gamma,y,i} \\ h(\alpha_k) \end{bmatrix} + \vec{\Delta}_{shift}(\alpha_k) + \vec{\Delta}_{mom} + \vec{\Delta}_{static}$ → Desired Drone Position

- $\mathbf{D}_{i,z}(\alpha_k) = \max\left(\mathbf{D}_{i,z}(\alpha_k),\ z_{min}\right)$ → Desired z Drone Position ($z_{min} \approx 1.5$)

6. **Cables & Tensions**:

a. $\vec{L}_i(\alpha_k) = \vec{\mathbf{D}}_i(\alpha_k) - (\vec{CoM} + \mathbf{R}_{des,p} \cdot \vec{att}_{eff,i})$ → Cable Length

b. $\hat{u}_{L_i}(\alpha_k) = \frac{\vec{L}_i(\alpha_k)}{\|\vec{L}_i(\alpha_k)\|}$ → Cable Length direction versor ($\mathbb{R}^3$)

c. $\vec{A}_i(\alpha_k) = \begin{bmatrix} \hat{u}_{L_i}(\alpha_k) \\ (\mathbf{R}_{des,p} \cdot \vec{att}_{eff,i}) \times \hat{u}_{L_i}(\alpha_k) \end{bmatrix}$ → Configuration Matrix i-th column ($\mathbb{R}^6$)

d. $\mathbf{A}(\alpha_k) = \left[\vec{A}_1(\alpha_k), \vec{A}_2(\alpha_k), \ldots, \vec{A}_N(\alpha_k)\right]$ → Configuration Matrix ($\mathbb{R}^{6\times N}$)

e. $\vec{T}^*(\alpha_k) = \arg\min_{\vec{T}} \frac{1}{2}\vec{T}^T\vec{T} \quad \text{s.t. Equality Constraints, Inequality Constraints}$ → Optimal Tensions Configuration ($\mathbb{R}^N$)

   a. $\min_{\vec{T}} \left\| \mathbf{A}(\alpha_k)\vec{T} - \vec{W}_{req} \right\|^2$ → Equality Constraints - Residual Minimization for Static Equilibrium Equation

   b. $\vec{T}_i(\alpha_k) \geq \vec{T}_{safe}$ → Inequality Constraints - Safety

   c. $\vec{T}_i(\alpha_k) \leq \vec{T}_{limit}$ → Inequality Constraints - Limit

h. $\vec{T}_{max}(\alpha_k) = \max(\vec{T}^*(\alpha_k))$ → Higher Tension
$\vec{T}_{min}(\alpha_k) = \min(\vec{T}^*(\alpha_k))$ → Lower Tension

7. Cost:

$J(\alpha_k) = w_T \cdot J_T(\alpha_k) + w_{barrier} \cdot J_{barrier}(\alpha_k) + w_{ref} \cdot J_{ref}(\alpha_k) + w_{smooth} \cdot J_{smooth}(\alpha_k) + w_{cond} \cdot J_{cond}(\alpha_k) + w_{resid,f} J_{resid,f}(\alpha_k) + w_{resid,m} J_{resid,m}(\alpha_k)$ → Optimization Function

1. $J_{cond}(\alpha_k) = \frac{\log_{10}(\kappa(\mathbf{A}(\alpha_k)))}{cond_{scale}}$ → Condition Matrix Cost
   - $\kappa(\mathbf{A}(\alpha_k)) = \min(\|\mathbf{A}(\alpha_k)\| \cdot \|\mathbf{A}(\alpha_k)^+\|, 1000)$ → Condition Number (how much the matrix is near to the singularity)
2. $J_{barrier}(\alpha_k) = \sum_{i=1}^{N} \left(B_{low,i}(\alpha_k) + B_{high,i}(\alpha_k)\right)$ → Barrier Cost
   - $B_{low,i}(\alpha_k)$ → Secuirty Low Margin
   - $B_{high,i}(\alpha_k)$ → Secuirty High Margin
3. $J_T(\alpha_k) = \frac{\vec{T}_{max}(\alpha_k)}{\vec{T}_{limit}}$ → Energy Cost
4. $J_{ref}(\alpha_k) = \frac{(\alpha_k - \alpha_{ref,dyn})^2}{\alpha_{scale}^2}$ → Normalized Reference Angle Cost
5. $J_{smooth}(\alpha_k) = \frac{(\alpha_k - \alpha_{prev})^2}{\alpha_{smooth}^2}$ → Normalized Smoothness Cost
6. $J_{resid}(\alpha_k) = \frac{1}{2}\|\mathbf{A}(\alpha_k)\vec{T}^*(\alpha_k) - \vec{W}_{req}\|^2$ → Static Equilibrium Error
   a. $J_{resid,f}(\alpha_k) = \frac{\sum_{j=1}^{3}((\mathbf{A}(\alpha_k)\vec{T}^*(\alpha_k))_j - W_{req,j})^2}{F_{nom}^2}$

b. $J_{resid,m}(\alpha_k) = \frac{\sum_{j=4}^{6}((\mathbf{A}(\alpha_k)\vec{T}^*(\alpha_k))_j - W_{req,j})^2}{M_{nom}^2}$

Final Geometry & Winch Damping

1. Calculate $\vec{\mathbf{D}}_i(\alpha_{opt})$, $\vec{L}_i(\alpha_{opt})$, $\vec{T}^*(\alpha_{opt})$
2. $\hat{u}_{L_i}(\alpha_{opt}) = \frac{\vec{L}_i(\alpha_{opt})}{\|\vec{L}_i(\alpha_{opt})\|}$ → Cable Length direction versor ($\mathbb{R}^3$)
3. $v_{cable,i} = (\vec{v}_{uav,i} - (\vec{v}_p + \vec{\omega}_p \times (\mathbf{R}_{des,p} \cdot \vec{att}_{eff,i})) \cdot \hat{u}_{L_i}$ → Relative Velocity Drone - Attachment Point
4. $L_{rigid,i} = \|\vec{L}_i(\alpha_{opt})\|$
5. $k_{stiff,i} = k_{cable} \cdot \left(\frac{L_{ref}}{L_{rigid,i}}\right)$ → Dynamic Cable Stiffness
6. $k_{damp,i} = 2 \cdot \zeta_{des} \cdot \sqrt{k_{stiff,i} \cdot \frac{m_{tot}}{N}}$ → Viscous Damping Coefficient ($\zeta_{des} \approx 0.7$)
7. $\Delta L_{static,i} = \frac{\vec{T}_i^*(\alpha_{opt})}{k_{stiff,i}}$ → Static Elastic Compensation
8. $\Delta L_{vel,i} = \frac{k_{damp,i} \cdot \vec{v}_{cable,i}}{k_{stiff,i}}$
9. $\dot{T}_{filt,i}^{(k)} = (1 - \beta_{jerk})\dot{T}_{filt,i}^{(k-1)} + \beta_{jerk}(\frac{T_i^{(k)} - T_i^{(k-1)}}{\Delta t})$ → Filtered Tension Variation ($\beta_{jerk} \approx 0.1$)
10. $\Delta L_{dynamic,i} = \tau_{relax} \cdot \frac{\dot{T}_{filt,i}}{k_{stiff,i}}$ → Feed-forward Jerk Compensation ($\tau_{relax} \approx 0.1$)
11. $\Delta L_{target,i} = \text{clamp}(\Delta L_{vel,i} + \Delta L_{dyn,i}, -L_{lim}, L_{lim})$ → Active Winch Damping Clamp ($L_{lim} \approx 0.5$)
12. $\delta_{smooth} = \text{clamp}(\Delta L_{target,i} - \Delta L_{corr,i}^{(k-1)}, -0.002, 0.002)$
13. $\Delta L_{corr,i} = \Delta L_{corr,i}^{(k-1)} + 0.2 \cdot \delta_{smooth}$
11. $L_{cmd,i} = \max(L_{rigid,i} - \Delta L_{static} - \Delta L_{corr}^{(k)}, L_{min})$ → Elastic Cable Length Command (with elastic compensation and Active Winch Damping)
12. $\vec{F}_{ff,i} = \vec{T}_i^*(\alpha_{opt}) \cdot \hat{u}_{L_i}(\alpha_{opt})$ → Feed-forward Force Vector for UAV Control